

IET, DAVV, Indore
B.E. II (ETC)
Class Test II (Oct 2023)
3ATR2C1: Applied Mathematics-III

Time: 70 min

Max Marks: 20

Note: Attempt any FOUR questions. All questions carry equal marks.

Q.1. Find the Taylor's and Laurent's series which represents the function $f(z) = \frac{(z-2)(z+2)}{(z+1)(z+4)}$, (05)

In the regions: (i) $|z| < 1$, (ii) $1 < |z| < 4$, (iii) $|z| > 4$.

Q.2. Find the residue of the function $f(z) = \frac{z^2 - 2z}{(z+1)^2(z^2+4)}$ at each pole. Using Residue theorem, (05)

evaluate $\int_C f(z) dz$, where $C: |z| = 1.5$.

Q.3. State Cauchy's integral theorem and evaluate the integral by using Cauchy's Integral (05)

Formula $\int_C \frac{4-3z}{z(z-1)(z-2)} dz$ where C is the circle $|z| = 3/2$.

Q.4. Evaluate $\int_0^1 \frac{dx}{1+x}$ using (i) Trapezoidal rule (ii) Weddle's rule. ($N=?$) (05)

Q.5. Find the value of $f(8)$ for the given set of discrete data. (05)

X:	4	5	7	10	11	13
F(x):	48	100	294	900	1210	2028

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